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Appendix

A Method Summary

A.1 Monolingual Human vs, Machine

Team Genaios_{STA_mono:1} (Sarvazyan et al., 2024) achieves the highest accuracy on Subtask A – Monolingual by extracting token-level probabilistic features using four Llama-2 models: Llama-2-7b, Llama-2-7b-chat, Llama-2-13b, and Llama-2-13b-chat. For each token they compute the log probability of the observed token, the log probability of the token predicted by each of the language models, and the entropy of the distribution. These features are then fed to a Transformer Encoder trained in a supervised fashion to detect synthetic text.

Team USTC-BUPT_{STA_mono:2} (Guo et al., 2024) incorporates domain adversarial neural networks into the task of machine-generated text detection to reach the second position in the ranking of Subtask A – Monolingual. They add a gradient reversal layer on top of the baseline, a supervised classifier based on RoBERTa. In addition, they exploit domain labels to enhance the transferability of learning between training and testing datasets. Their architecture is based on RoBERTa and adds two classification heads, one for category classification (human or synthetic) and one for domain classification (e.g. news, essays, etc.), the former uses an MLP layer and the latter is composed of an MLP together with a gradient reversal layer. Finally, the loss is also adapted by summing together the category and the domain losses. The submission evaluated an improvement of approximately 8% compared to the baseline.

Team PetKaz_{STA_mono:12} (Petukhova et al., 2024) uses a PLM, RoBERTa-base, fine-tuned for synthetic text detection and enhances it with linguistic features, to train a feed-forward binary classifier (human or synthetic). Their final model uses diverse features and notably, they undersample the human data.

Team HU_{STA_mono:17} (Roy Dipta and Shahriar, 2024) Adopts an architecture trained with a contrastive learning approach based on fine-tuning *sentence-transformers/all-mpnet-base-v2*. The model is trained on an augmented dataset obtained by paraphrasing sentences in the training set.

Team TrustAI_{STA_mono:20} (Urlana et al., 2024) tries two approaches: (a) an ensemble approach with the combination of Multinomial Naive Bayes, LGBM Classifier (lightGBM classifier) and SGD classifier. Each is trained on the concatenation of tf-idf and spaCy embeddings obtained from the Subtask A – Monolingual dataset and (b) a synthetic text classifier based on RoBERTa fine-tuned first with the outputs of the 1.5B-parameter GPT-2 model and subsequently on the Subtask A – monolingual dataset. They show that exploring methodologies with different assumptions helps identify the best performing approach.

Team RFBES_{STA_mono:23} (Heydari Rad et al., 2024) Both semantic and syntactic considerations were taken into account. For semantic analysis, emphasis was placed on smaller text segments rather than the entire document, operating under the belief that AI models could produce similarly coherent long texts as humans. To achieve this, the XLM-RoBERTa model was employed. Regarding syntactic analysis, a stacked bidirectional LSTM model was used to categorize texts based on their grammatical patterns using UPOS tags. Interestingly, no significant differences in UPOS tag distribution between AI-generated and human-written texts were revealed by the findings.

Team L3i++_{STA_mono:24} (Tran et al., 2024) Proposes a comparative study among 3 groups of methods to detect synthetic texts: 5 likelihood-based methods; 2 fine-tuned sequence-labeling language models (RoBERTa, XLM-RoBERTa); and a fine-tuned large language model, llama-2-7b. LLaMA 2 outperforms the rest and accurately detects machine-generated texts.

Team art-nat-HH_{STA_mono:25} (Ciccarelli et al., 2024) fine-tunes a RoBERTa model pre-trained for AI-detection and combines it with a set of linguistic features: syntactic, lexical, probabilistic and stylistic. To improve the classifier, they train two separate neural networks on these features, one for each class predicted by the RoBERTa-based classifier.

Team Unibuc - NLP_{STA_mono:28} (Marchitan et al., 2024) fine-tunes a Transformer-based model with a MLP as a classification head. They combine the datasets of Subtask A – monolingual and Subtask B to obtain a larger training set.

NewbieML_{STA_mono:30} (Tran and Tran, 2024) embeds texts with Longformer-large. Then they Ensemble SVM, LogisticRegression and XGBoost with, as a meta model, a KNN.

Team QUST_{STA_mono:31} (Xu et al., 2024) experiments with multiple models on a dataset extended through data augmentation. They select the two best-performing models for ensembling: (1) a fine-tuned RoBERTa model, combined with the Multiscale Positive-Unlabeled (MPU) training and (2) a DeBERTa model. They use these two for model fusion through stacking ensemble.

Team NootNoot_{STA_mono:39} (Bahad et al., 2024) carefully fine-tunes a RoBERTa-base model to classify human written and synthetic texts.

Team Mast Kalandar_{STA_mono:40} (Bafna et al., 2024) trains a classifier that uses a frozen RoBERTa model with an LSTM head to classify human vs machine written texts.

Team I2C-Huelva_{STA_mono:41} (rodero et al., 2024) proposes a method to use multimodal models together with text analysis to enhance synthetic text detection. To mix the two approaches they explore ensemble by testing several voting methods.

Team Werkzeug_{STA_mono:45} (Wu et al., 2024) uses Roberta-large and XLM-roberta-large to encode texts. To address the anisotropic embedding space created by transformer-based language models, they employ several learnable parametric whitening (PW) transformation. They show that addressing the anisotropy of the embedding space improves accuracy in detecting synthetic text.

Team NCL-UoR_{STA_mono:50} (Xiong et al., 2024) fine-tunes several PLMs including XLM-RoBERTa, RoBERTa with Low-Rank Adaptation (LoRA) and DistilBERT. Finally, they use majority voting ensembling with XLM-RoBERTa and LoRA-RoBERTa. To confirm that ensembling is a strong technique to boost synthetic text classification accuracy.

Team Sharif-MGTD_{STA_mono:51} (Ebrahimi et al., 2024) carefully fine-tunes RoBERTa-base for synthetic text detection, show that pre-trained language models are a versatile approach.

Team BadRock_{STA_mono:*} (Siino, 2024) is based on a fine-tuning of a DistilBERT trained on the SST-2 dataset.

Team Collectivized Semantics_{STA_mono:62} (Datta et al., 2024) fine-tunes Roberta-base using Ada-LoRA and uses the weighted sum of all the layer hidden states' mean as features to train a classifier. They show that exploiting the knowledge at all layers of encoder language models helps when detecting synthetic texts.

Team IUCL_{STA_mono:63} (Pulipaka et al., 2024) tries both classical ML classifiers, Naive Bayes and Decision Trees as well as fine-tuning transformers and they conclude that fine-tuned RoBERTa is best among the methods they try.

Team SINAI_{STA_mono:67} (Gutiérrez Megías et al., 2024) compares three methods: (a) supervised classification, based on fine-tuning the XLM-RoBERTa-Large language model; (b) likelihood-based methods, using GPT-2 to compute the perplexity of each text and use this perplexity as a score; (c) a hybrid approach that merges text with its perplexity value into a classification head. The choice of a mixed approach proves effective in improving synthetic text detection accuracy.

Team MasonTigers_{STA_mono:71} (Puspo et al., 2024) experiments with different transformer-based models: Roberta, DistilBERT, ELECTRA and ensembles these models. They also experiment with zero-shot prompting and finetuning FlanT5. Further confirming that ensembling is a strong methodology for detecting synthetic texts.